

Alpha Diversity

Muluh

Community diversity

Group-wise comparisons

- **Index:** shannon
- **Group variable:** meal_group
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

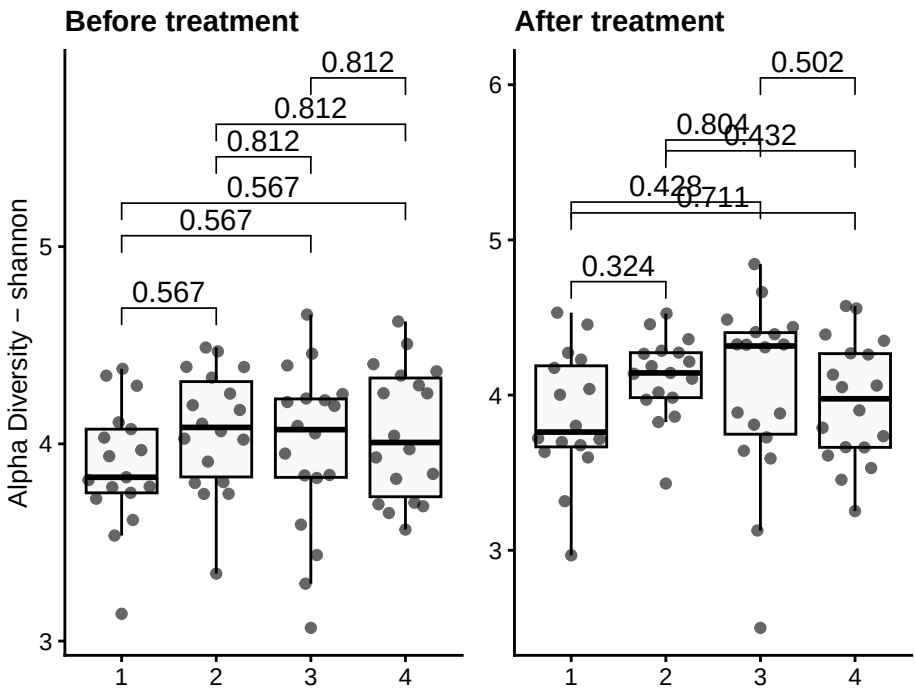


Table of the top findings

Across-group

Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

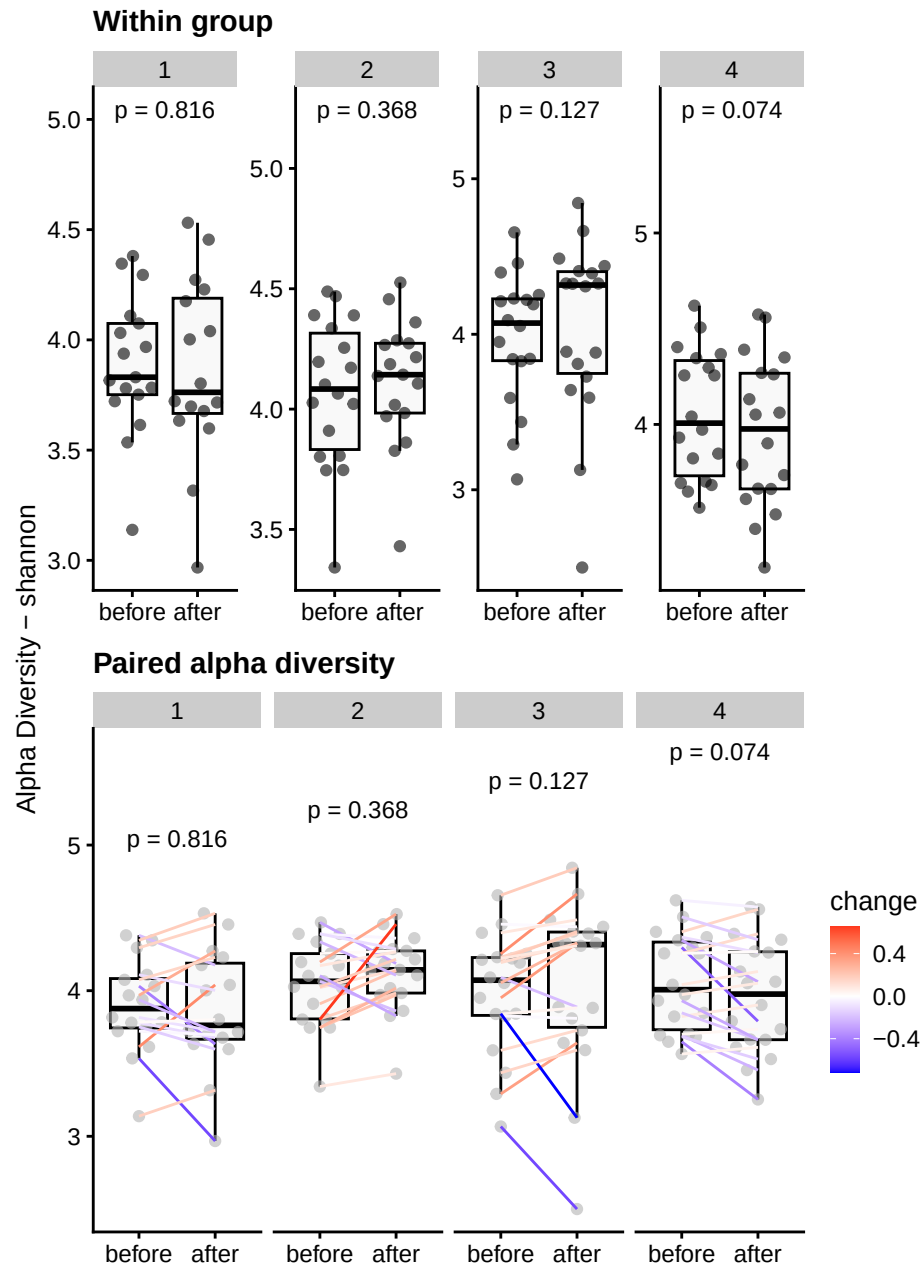
group1	group2	mean1	mean2	log2FC	p.adj
1	2	3.89	4.07	0.07	0.567
1	3	3.89	3.98	0.03	0.567
1	4	3.89	4.05	0.06	0.567
2	3	4.07	3.98	-0.03	0.812
2	4	4.07	4.05	-0.01	0.812
3	4	3.98	4.05	0.03	0.812

After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	3.86	4.12	0.09	0.324
1	3	3.86	4.04	0.06	0.428
1	4	3.86	3.96	0.03	0.711
2	3	4.12	4.04	-0.03	0.804
2	4	4.12	3.96	-0.06	0.432
3	4	4.04	3.96	-0.03	0.502



Within-group

Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	3.6	3.1, 4.0	<0.001	3.5	2.9, 4.1	<0.001
meal_group			0.3			0.3
1	—	—		—	—	
2	0.21	-0.03, 0.44	0.084	0.28	-0.02, 0.58	0.065
3	0.12	-0.12, 0.35	0.3	0.20	-0.09, 0.50	0.2
4	0.17	-0.06, 0.40	0.2	0.10	-0.20, 0.39	0.5
age	0.01	0.00, 0.01	0.2	0.01	0.00, 0.02	0.2

¹CI = Confidence Interval

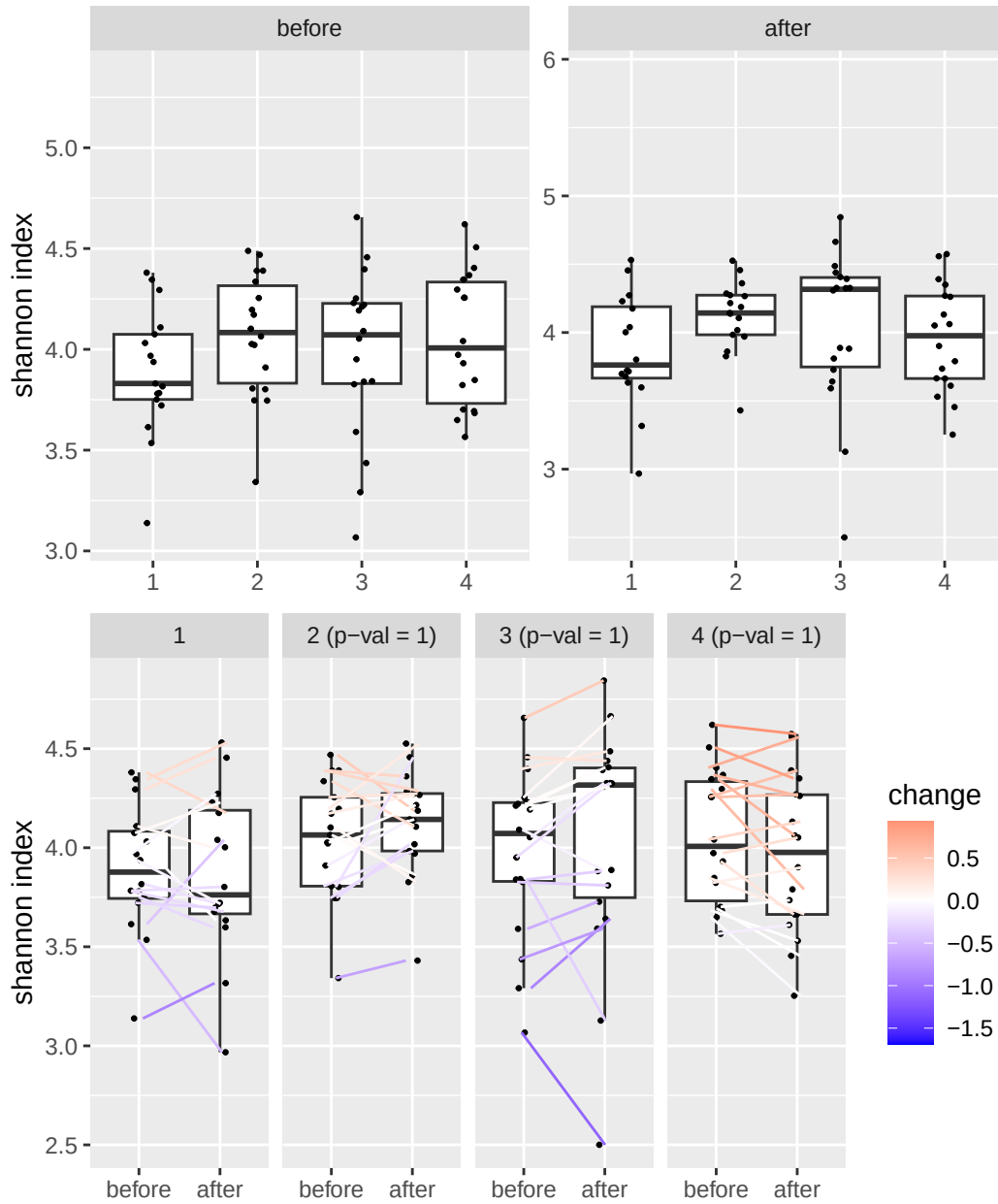
Paired test of before and after	Beta	95% CI ¹	p-value
(Intercept)	3.5	-Inf, Inf	<0.001
meal_group			0.4
1	—	—	
2	0.23	-Inf, Inf	>0.9
3	0.16	-Inf, Inf	>0.9
4	0.13	-Inf, Inf	>0.9
age	0.01	0.00, 0.01	0.039

¹CI = Confidence Interval

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

meal_group	mean1	mean2	log2FC	p.value
1	3.89	3.86	-0.01	0.816
2	4.07	4.12	0.03	0.368
3	3.98	4.04	0.02	0.127
4	4.05	3.96	-0.03	0.074

Alternate analyses



Data Summary

class: TreeSummarizedExperiment
dim: 1700 140

```

metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```

Version info

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R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS

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Matrix products: default
BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0

```

```

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

```

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time zone: Europe/Helsinki
tzcode source: system (glibc)

```

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attached base packages:
[1] stats4      grid        stats      graphics  grDevices  utils      datasets
[8] methods     base

```

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other attached packages:
[1] kableExtra_1.4.0      vegan_2.6-8
[3] lattice_0.22-6        permute_0.9-7

```

[5]	lubridate_1.9.3	forcats_1.0.0
[7]	stringr_1.5.1	purrr_1.0.2
[9]	readr_2.1.5	tibble_3.2.1
[11]	tidyverse_2.0.0	tidyr_1.3.1
[13]	sechm_1.13.1	ComplexHeatmap_2.21.1
[15]	scater_1.33.4	scuttle_1.15.4
[17]	quarto_1.4.4	patchwork_1.3.0
[19]	parameters_0.24.0	multtest_2.61.0
[21]	miaTime_0.1.21	miaViz_1.15.4
[23]	ggraph_2.2.1	mia_1.15.6
[25]	TreeSummarizedExperiment_2.13.0	Biostrings_2.73.2
[27]	XVector_0.45.0	SingleCellExperiment_1.27.2
[29]	MultiAssayExperiment_1.31.5	SummarizedExperiment_1.35.3
[31]	Biobase_2.65.1	GenomicRanges_1.57.1
[33]	GenomeInfoDb_1.41.2	IRanges_2.39.2
[35]	S4Vectors_0.43.2	BiocGenerics_0.51.3
[37]	MatrixGenerics_1.17.0	matrixStats_1.4.1
[39]	gridExtra_2.3	gtsummary_2.0.4
[41]	ggsignif_0.6.4	ggpubr_0.6.0
[43]	ggplot2_3.5.1	DT_0.33
[45]	dplyr_1.1.4	cardx_0.2.2
[47]	ANCOMBC_2.8.0	

loaded via a namespace (and not attached):

[1]	randomcoloR_1.1.0.1	gld_2.6.6
[3]	nnet_7.3-19	TH.data_1.1-2
[5]	vctrs_0.6.5	energy_1.7-12
[7]	digest_0.6.37	png_0.1-8
[9]	shape_1.4.6.1	proxy_0.4-27
[11]	Exact_3.3	registry_0.5-1
[13]	rbiom_1.0.3	ggrepel_0.9.6
[15]	bayestestR_0.15.0	parallelly_1.38.0
[17]	mediation_4.5.0	MASS_7.3-61
[19]	reshape2_1.4.4	foreach_1.5.2
[21]	withr_3.0.2	xfun_0.49
[23]	ggfun_0.1.6	survival_3.7-0
[25]	doRNG_1.8.6	commonmark_1.9.1
[27]	memoise_2.0.1	ggbeeswarm_0.7.2
[29]	emmeans_1.10.4	systemfonts_1.1.0
[31]	gmp_0.7-5	tidytree_0.4.6
[33]	zoo_1.8-12	GlobalOptions_0.1.2
[35]	gtools_3.9.5	V8_5.0.1
[37]	Formula_1.2-5	datawizard_0.13.0

[39] httr_1.4.7	rstatix_0.7.2
[41] globals_0.16.3	ps_1.8.0
[43] rstudioapi_0.17.1	UCSC.utils_1.1.0
[45] generics_0.1.3	base64enc_0.1-3
[47] processx_3.8.4	curl_5.2.3
[49] zlibbioc_1.51.1	ScaledMatrix_1.13.0
[51] polyclip_1.10-7	ca_0.71.1
[53] GenomeInfoDbData_1.2.13	SparseArray_1.5.41
[55] xtable_1.8-4	doParallel_1.0.17
[57] evaluate_1.0.1	S4Arrays_1.5.10
[59] hms_1.1.3	irlba_2.3.5.1
[61] labelled_2.13.0	colorspace_2.1-1
[63] readxl_1.4.3	magrittr_2.0.3
[65] later_1.3.2	viridis_0.6.5
[67] ggtree_3.13.1	DECIPHER_3.1.4
[69] cowplot_1.1.3	class_7.3-22
[71] Hmisc_5.2-0	pillar_1.9.0
[73] nlme_3.1-166	iterators_1.0.14
[75] decontam_1.25.0	compiler_4.4.2
[77] beachmat_2.21.6	stringi_1.8.4
[79] DescTools_0.99.57	TSP_1.2-4
[81] tokenizers_0.3.0	minqa_1.2.8
[83] plyr_1.8.9	crayon_1.5.3
[85] abind_1.4-8	gridGraphics_0.5-1
[87] haven_2.5.4	graphlayouts_1.2.0
[89] bit_4.5.0	rootSolve_1.8.2.4
[91] sandwich_3.1-1	codetools_0.2-20
[93] multcomp_1.4-26	BiocSingular_1.21.4
[95] slam_0.1-54	e1071_1.7-16
[97] lmom_3.2	GetoptLong_1.0.5
[99] tidytext_0.4.2	splines_4.4.2
[101] markdown_1.13	circlize_0.4.16
[103] Rcpp_1.0.13-1	sparseMatrixStats_1.17.2
[105] cellranger_1.1.0	knitr_1.48
[107] utf8_1.2.4	clue_0.3-65
[109] lme4_1.1-35.5	fs_1.6.5
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[113] DelayedMatrixStats_1.27.3	Rdpack_2.6.1
[115] expm_1.0-0	gsl_2.1-8
[117] ggplotify_0.1.2	estimability_1.5.1
[119] Matrix_1.7-0	svglite_2.1.3
[121] tzdb_0.4.0	lpSolve_5.6.21
[123] tweenr_2.0.3	pkgconfig_2.0.3

[125]	tools_4.4.2	cachem_1.1.0
[127]	rbibutils_2.2.16	viridisLite_0.4.2
[129]	DBI_1.2.3	numDeriv_2016.8-1.1
[131]	gt_0.11.1	fastmap_1.2.0
[133]	rmarkdown_2.29	scales_1.3.0
[135]	cards_0.4.0	broom_1.0.7
[137]	coda_0.19-4.1	insight_1.0.0
[139]	carData_3.0-5	rpart_4.1.23
[141]	farver_2.1.2	tidygraph_1.3.1
[143]	mgcv_1.9-1	yaml_2.3.10
[145]	foreign_0.8-87	cli_3.6.3
[147]	lifecycle_1.0.4	mvtnorm_1.3-2
[149]	bluster_1.15.1	backports_1.5.0
[151]	BiocParallel_1.39.0	timechange_0.3.0
[153]	gtable_0.3.6	rjson_0.2.23
[155]	parallel_4.4.2	ape_5.8
[157]	SnowballC_0.7.1	CVXR_1.0-15
[159]	jsonlite_1.8.9	seriation_1.5.6
[161]	bit64_4.5.2	Rtsne_0.17
[163]	yulab.utils_0.1.8	BiocNeighbors_1.99.1
[165]	RcppParallel_5.1.9	janeaustrinr_1.0.0
[167]	lazyeval_0.2.2	htmltools_0.5.8.1
[169]	tinytex_0.54	glue_1.8.0
[171]	broom.helpers_1.17.0	treeio_1.29.1
[173]	boot_1.3-31	igraph_2.1.1
[175]	R6_2.5.1	broom.mixed_0.2.9.6
[177]	labeling_0.4.3	Rmpfr_1.0-0
[179]	cluster_2.1.8	rngtools_1.5.2
[181]	aplot_0.2.3	nloptr_2.1.1
[183]	DirichletMultinomial_1.47.0	DelayedArray_0.31.13
[185]	tidyselect_1.2.1	vipor_0.4.7
[187]	htmlTable_2.4.3	xml2_1.3.6
[189]	ggforce_0.4.2	car_3.1-3
[191]	future_1.34.0	rsvd_1.0.5
[193]	munsell_0.5.1	furrr_0.3.1
[195]	data.table_1.16.2	htmlwidgets_1.6.4
[197]	RColorBrewer_1.1-3	rlang_1.1.4
[199]	lmerTest_3.1-3	ggnewscale_0.5.0
[201]	fansi_1.0.6	beeswarm_0.4.0